ddqc: Biology-inspired data-driven quality control for scientific discovery in single-cell transcriptomics

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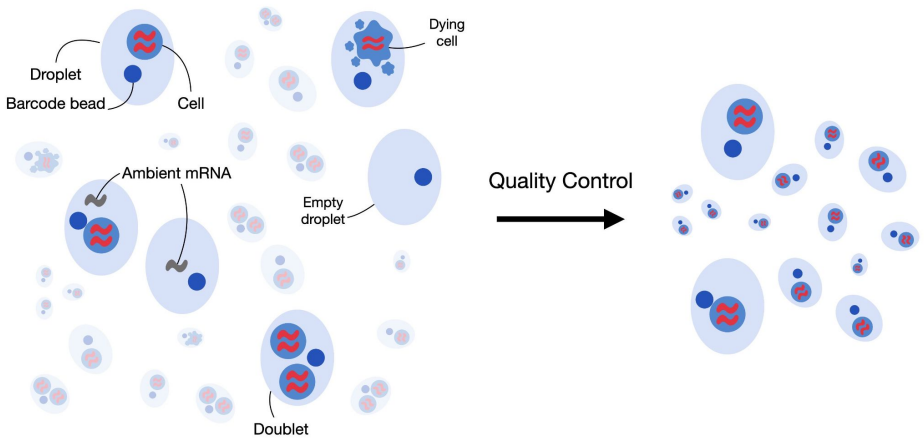
Introduction

Why Quality Control Matters in scRNA-seq



QC:
Filtering out low-quality or non-biological cells using:
- Gene counts (nGenes)
- Total counts (nCounts)
- % mitochondrial reads

Poor QC:
- Low gene counts → empty droplets
- High mitochondrial % → stressed/dying cells
- Doublets/multiplets → mixed cell signals

Impact of poor QC:
- Noisy data
- Blurred or false clusters
- Misleading biological conclusions

Standard QC uses global thresholds → ignores biological variation

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Biology-inspired data-driven quality control for scientific discovery in single-cell transcriptomics

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Limitations of Standard QC

Standard QC:

- Used fixed global thresholds:
  - nGenes (e.g., <200 or >5,000)
  - % mitochondrial (e.g., >5–20%)
- Applied globally to all cells
- Removes valid cells
  - e.g., activated or stressed cells with high mitochondrial %
- Misses low-quality cells
- Dataset-dependent

Biology-Inspired Data-Driven QC (ddqc)

No QC:

- Raw data (baseline)
- Shows impact of not filtering

Standard QC (global thresholds):

- Same thresholds applied to all cells
- Filters using fixed cutoffs:
  - nGenes
  - nCounts
  - % mitochondrial

miQC (Probabilistic QC):

- Fits a mixture model using QC metrics
- Assigns each cell a probability of being low-quality
- Filters based on that probability
- **Global method (not cluster-aware)**

ddqc (proposed method):

- **Cluster-aware QC**
- Cluster cells to capture biological structure
- Perform QC within clusters (not globally)
- Use MAD-based thresholds per cluster
- Removes cells that are outliers relative to similar cells

Method Evaluation:

1) **Clustering structure (UMAP)**
   - Compare how clean and separable clusters look
   - Assess whether biological structure is preserved
2) **Cell retention**
   - Number of cells kept vs removed
   - Check if methods are too aggressive or too lenient
3) **Biological interpretability**
   - Do clusters correspond to known cell types?
   - Are marker genes preserved?
   - Are meaningful populations retained?
4) **Noise vs signal**
   - Whether low-quality cells are effectively removed
   - Whether true biological variation is preserved

Reproduction + Extensions

Reproduction

(Colon Immune Dataset – Smillie et al., 2019):

- Compared 4 QC strategies:
  - No QC (baseline)
  - Standard cutoff (nGenes < 200, mito > 10%)
  - miQC (probabilistic mixture model)
  - ddqc (cluster-based, MAD thresholds)
- Started from the same unfiltered dataset (~210k cells)
- Used identical Seurat pipeline (normalization → PCA → clustering → UMAP)

Goal: isolate how QC choice affects clustering structure and cell retention

Extensions

(1) Disease Context (Kidney: Healthy vs IgAN):

- Test robustness when QC metrics shift due to biology
- Compare ddqc vs miQC retention and filtering behavior

(2) Clustering Resolution Sensitivity (ddqc):

- Vary resolution: 0.4 → 2.0
- Evaluate impact on clustering structure and cells removed

(3) Cluster Coherence Validation:

- Compare clusters to CellTypist + SingleR annotations
- Assess whether clusters reflect biological cell types

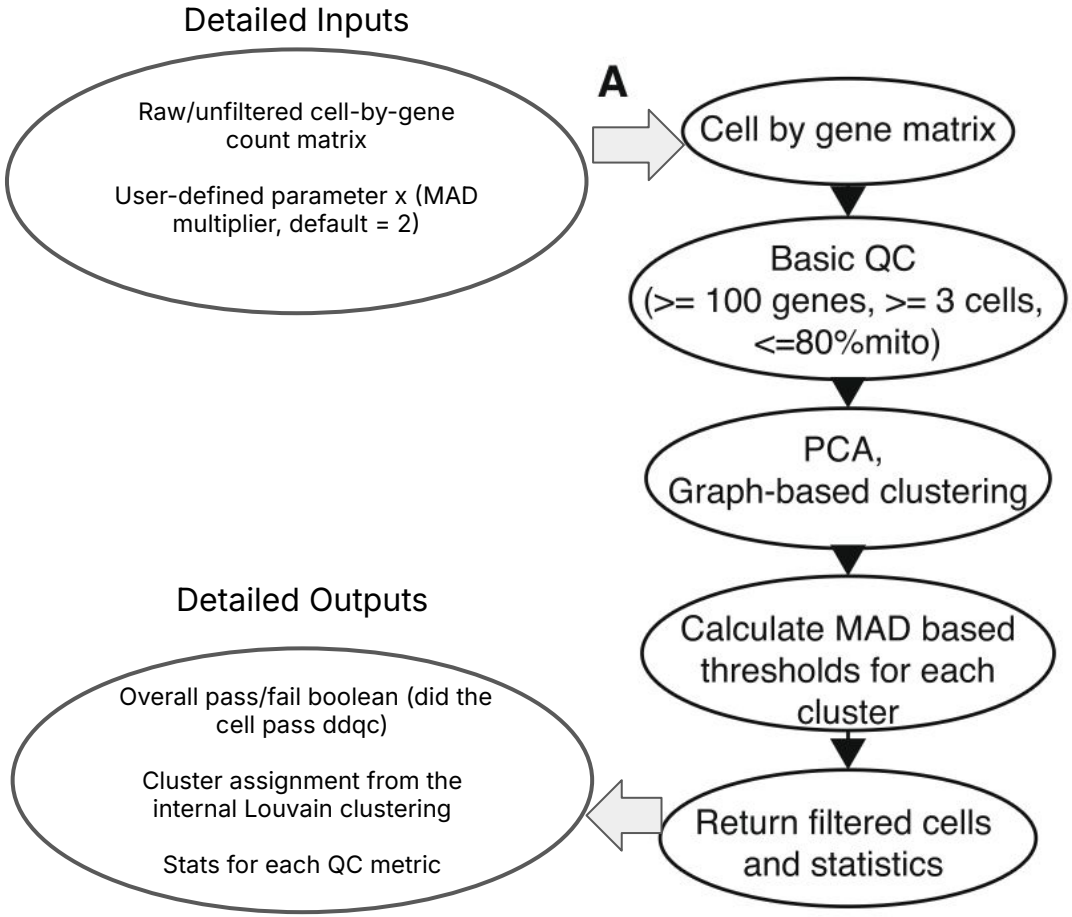
Methods

ddqc Core Idea

Assumption: cells within a cluster share the same or closely related cell type, so their QC metric distributions reflect biology, not noise

- Instead of one global threshold, compute thresholds per cluster
- Outliers within a cluster are more likely to be truly low-quality relative to their peers

ddqc Pipeline



Reproduction

Reproduction Objective

This study reproduces key findings from Subramanian et al. (2022), which compare multiple quality control (QC) strategies in single-cell RNA sequencing data.

Four QC approaches that we evaluated:

- no quality control (no QC)
- standard threshold-based filtering (standard cutoff)
- miQC (mixture model-based quality control)
- ddqc (data-driven quality control)

The primary objective was to assess how differently the QC strategies influence cell retention and clustering structure, as visualized through UMAP embeddings.

Reproduction Strategy

The analysis was performed on the colon immune cell dataset used in the original study.

All methods used the same unfiltered Seurat object, ensuring consistency in the input data. The methods differed in their filtering criteria, but the downstream analysis pipeline was kept identical across all methods.

Following the QC, each dataset underwent:

- Normalization and log-transformation
- Selection of 2,000 highly variable genes
- Data scaling
- Principal component analysis with top 20 PCs
- Louvain clustering with resolution of 1.4
- UMAP visualization

This standardized workflow was useful in observing differences in the clustering structure or cell retention. It helped attribute these differences solely due to the choice of the QC method.

Cell Retention Across QC Methods

Initial dataset: **210,614 cells**

- **ddqc:**
  - 210,066 cells retained (~99.7%)
- **miQC:**
  - 196,050 cells retained (~93%)
- **Standard cut-off:**
  - 198,586 cells retained (~94%)
- **No QC:**
  - No filtering applied

- Approximate cells removed:
  - ddqc: ~600 cells
  - miQC: ~14,000 cells
  - standard cut-off: ~12,000 cells

These results indicate a substantial difference in the number of cells that are retained for each method, highlighting the stringency of filtering. ddqc retains nearly the entire dataset, removing only a small fraction of cells, whereas both miQC and standard cut-off filtering remove a significantly larger number of cells.

This pattern is consistent with the findings reported in the original study, where ddqc is designed to preserve a broader range of cells by applying clutter-specific thresholds, rather than relying on global filtering methods.

UMAP Comparison Across QC Methods

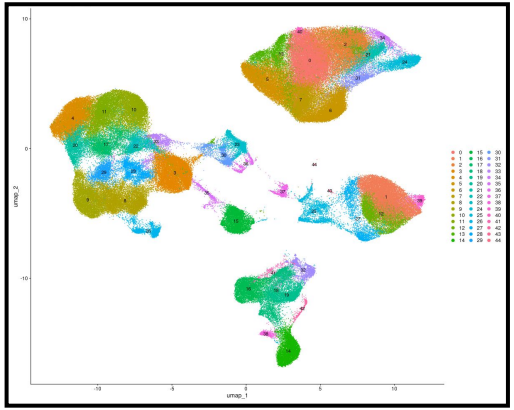


Fig 3. UMAP from no QC processing

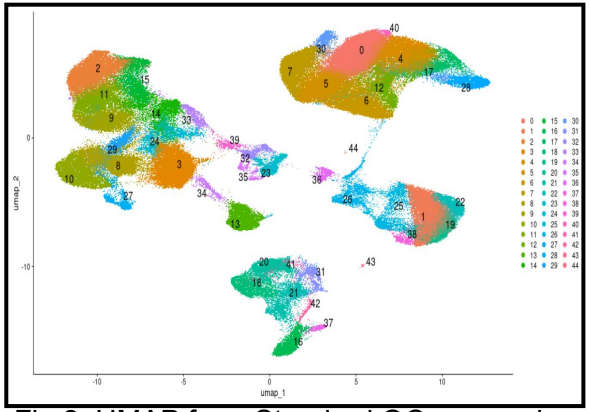


Fig 2. UMAP from Standard QC processing

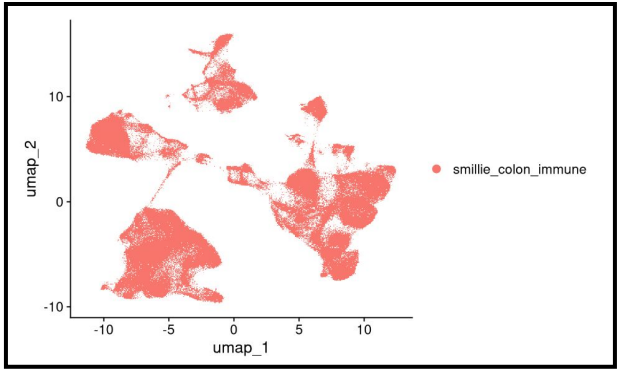


Fig 4. UMAP plot produced from miQC preprocessing

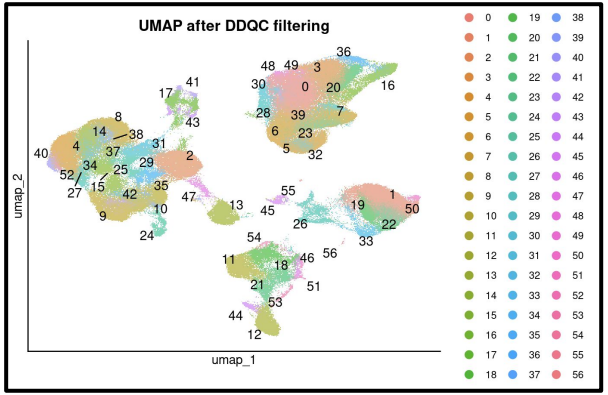


Fig 1. UMAP plot produced from ddqc preprocessing

No QC:

- Diffuse and overlapping clusters
- High noise and poor separation

Standard QC:

- Improved separation
- Some fragmentation and isolated clusters

miQC:

- Strong filtering of low-quality cells
- Cleaner clusters but reduced cell density

ddqc:

- Compact and well-defined clusters
- Preserves overall structure and cell diversity

Reproduction Outcome

Important insights:

- ddqc retains the highest proportion of cells (~99.7%)
- miQC and standard QC remove a larger fraction of cells
- No QC results in noisy and poorly defined clusters

Overall, our results are consistent with the findings reported in Subramanian et al. (2022). ddqc preserves a larger proportion of cells while improving clustering structure compared to no QC and standard QC approaches.

While miQC produces cleaner clusters through more aggressive filtering, it removes a substantial number of cells, which may include biologically relevant populations.

These results demonstrate that ddqc achieves a balance between noise reduction and preservation of biological variation, successfully reproducing the key trends observed in the original study.

Critiques

Critiques

- The paper assumed cells clustered together has biological relevance or share the same biology
- The paper did not run ddqc with disease vs healthy dataset
- The paper did not explore different clustering resolutions used in ddqc

Extensions

Exploring miQC and ddqc differences on healthy vs disease dataset (kidney)

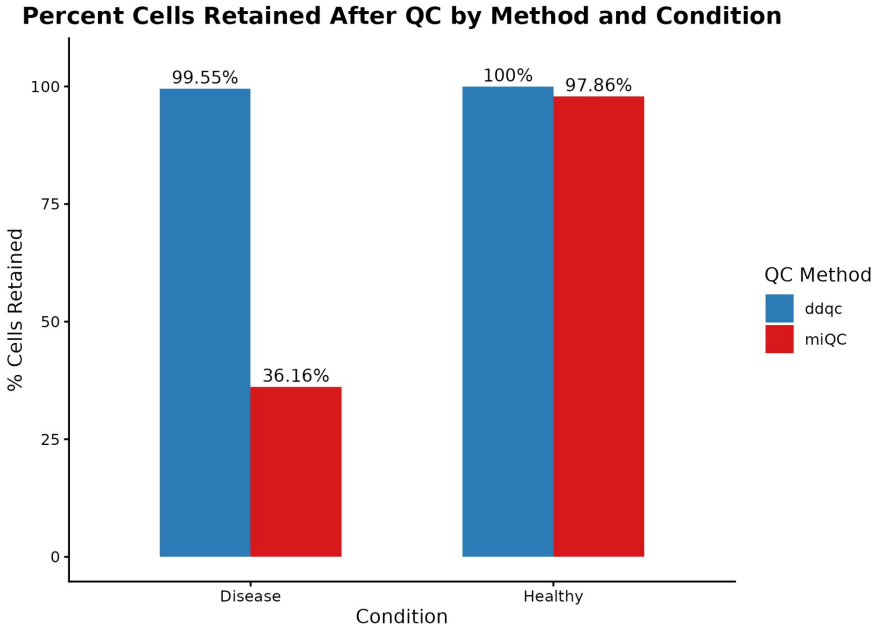


Fig 5. A bar chart showing the percentage retention rate of cells between the healthy and disease state using ddqc and miQC

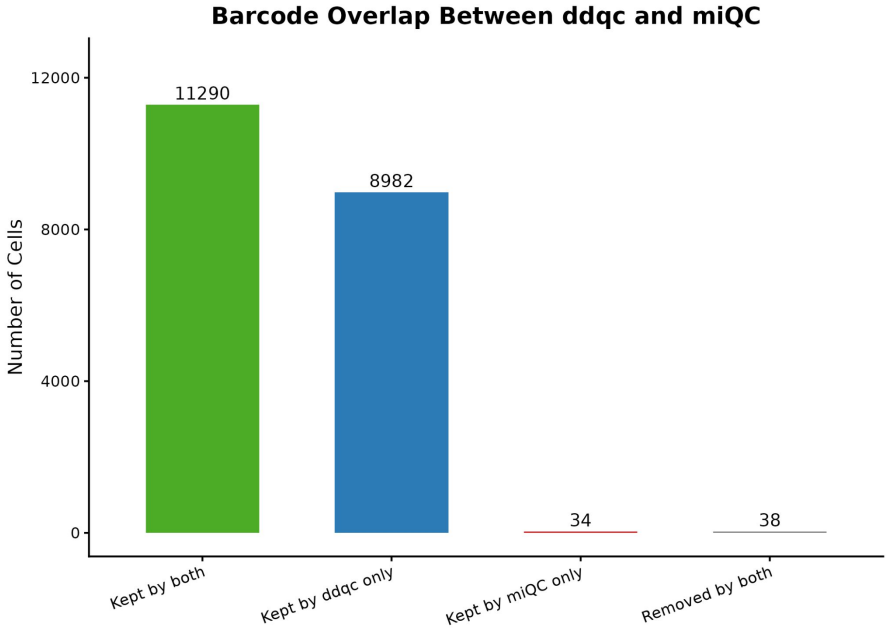


Fig 6. A bar chart showing the overlapping cell barcodes between ddqc and miQC

Exploring miQC and ddqc differences on healthy vs disease dataset (kidney)

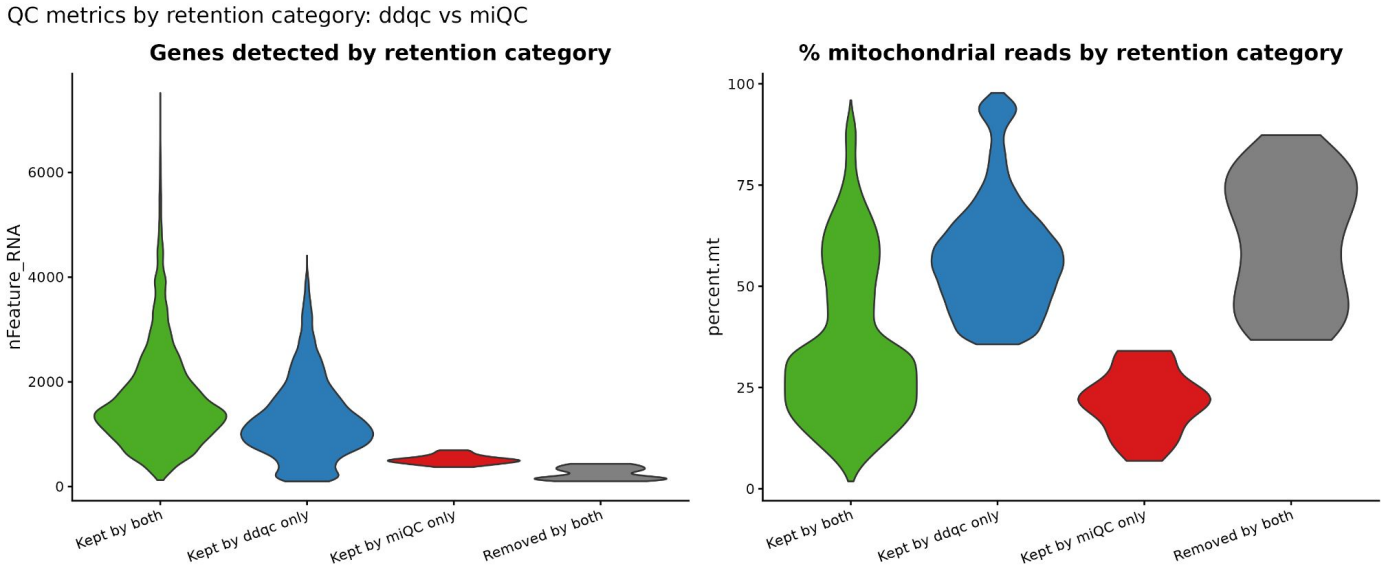


Fig 7. violin plots illustrating the number of genes detected by retention and % mitochondrial reads retention between ddqc and miQC

Exploring miQC and ddqc differences on healthy vs disease dataset (kidney)

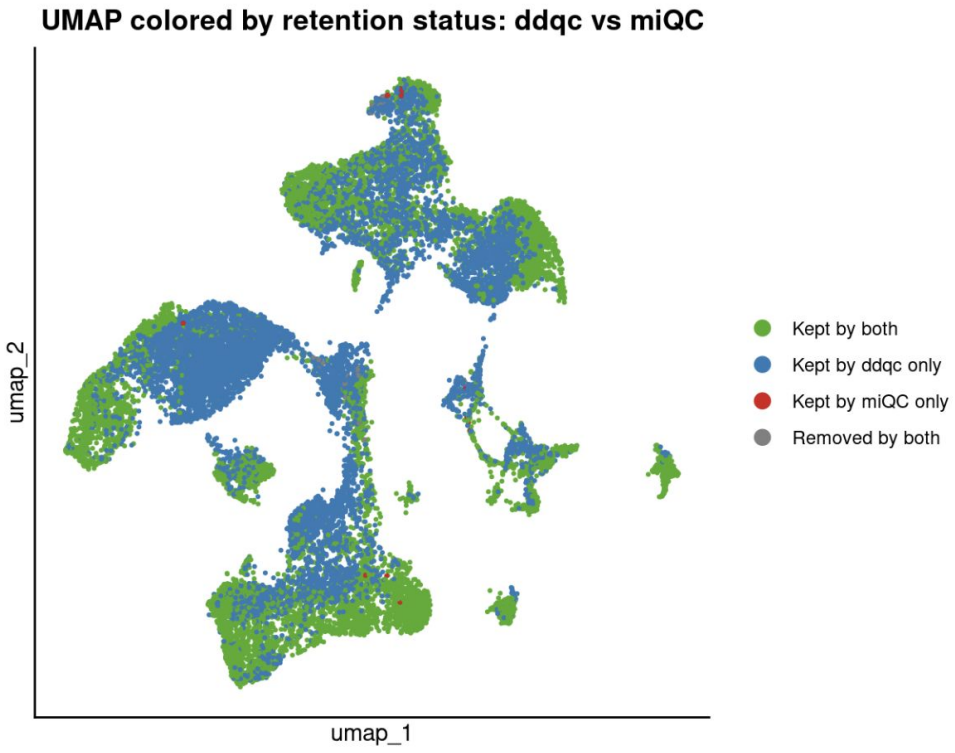


Fig 8. UMAP colored by retention status between ddqc and miQC

Exploring miQC and ddqc differences on healthy vs disease dataset (kidney)

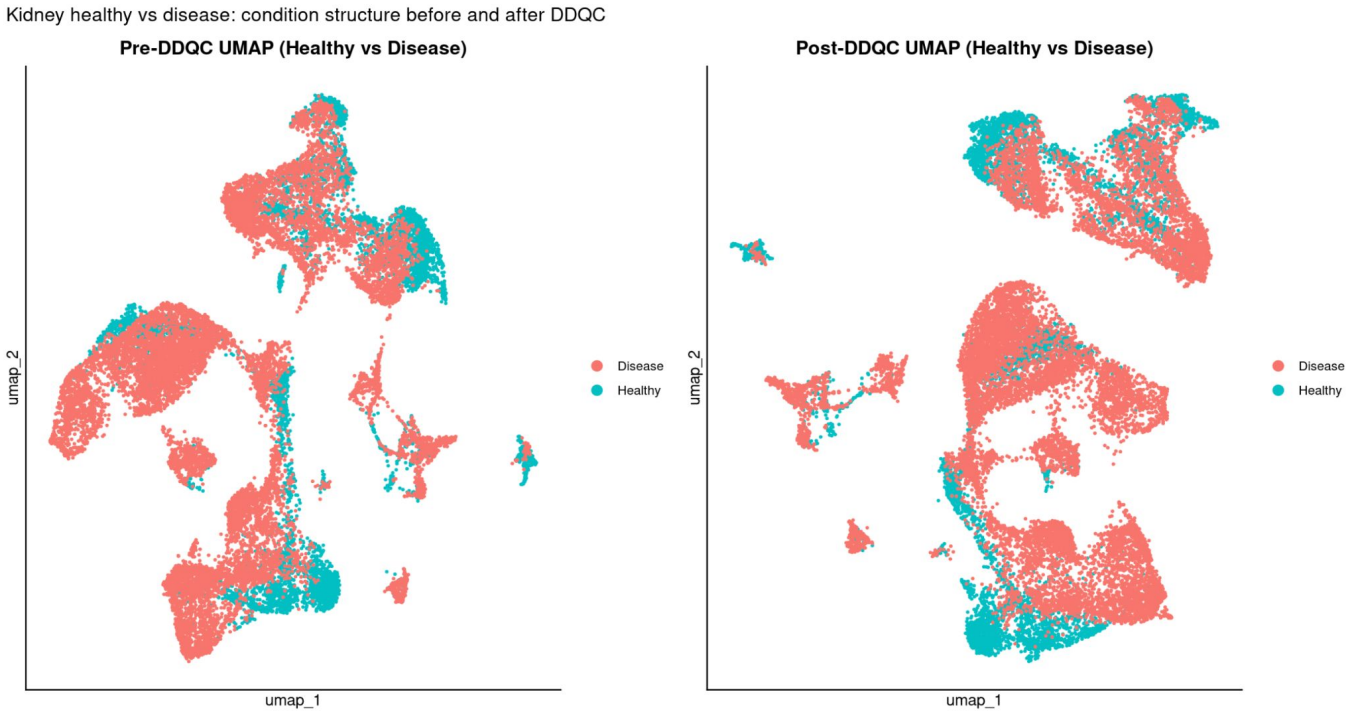
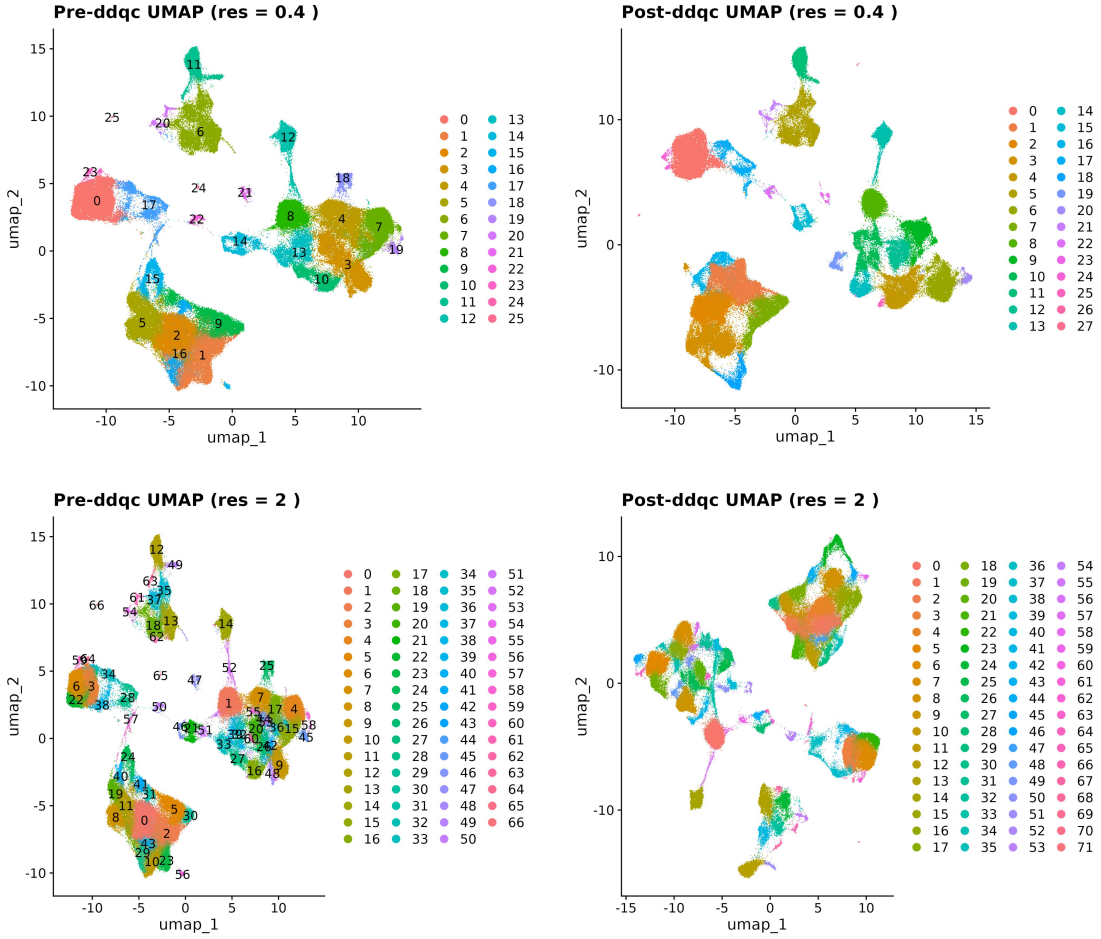


Fig 9. UMAP visualization of cell distribution and clustering structure in disease vs. healthy samples before and after ddqc

Fig 10. UMAP before and after ddqc at low (0.4) and high (2.0) clustering resolution



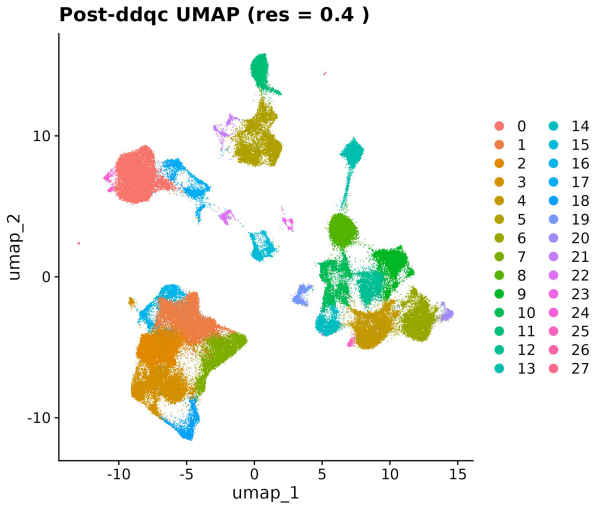
Experimenting with the resolution parameter for clustering

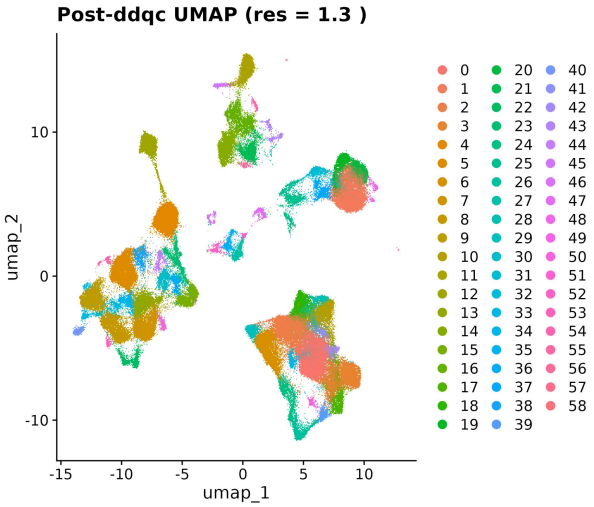
Effect of ddqc Across Clustering Resolutions:

- **Pre-ddqc:** diffuse clusters, noisy boundaries, scattered cells
- **Post-ddqc:** tighter clusters, clearer separation

Improvement is consistent at:

- Low resolution (0.4) → cleaner broad populations
- High resolution (2.0) → reduced noise but still fragmented





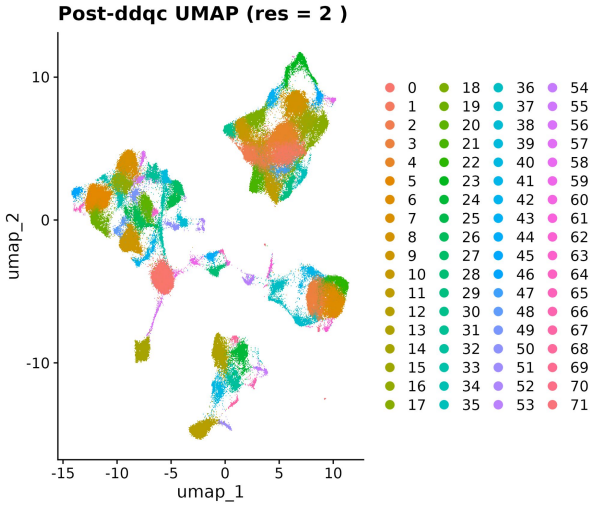


Fig 11. Post-ddqc UMAP across resolutions (0.4, 1.3, 2.0)

Experimenting with the resolution parameter for clustering

Effect of Resolution on Clustering After ddqc:

Resolution controls cluster granularity

- 0.4 → broad, merged clusters
- 1.3 → balanced structure
- 2.0 → fragmented clusters

ddqc improves data quality, but does not eliminate sensitivity to resolution

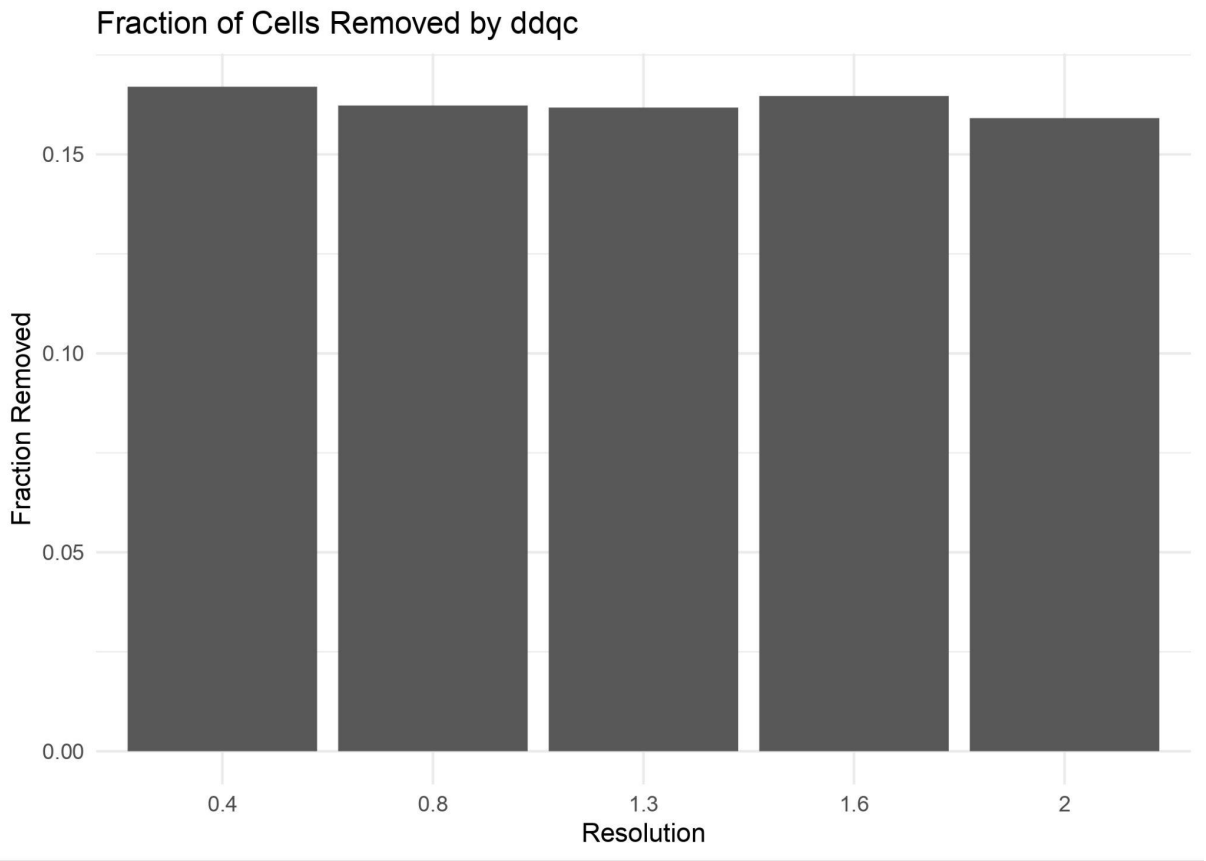


Fig. Fraction of Cells Removed by ddqc Across Clustering Resolutions

Experimenting with the resolution parameter for clustering

- ~16–17% of cells removed across all resolutions
- Variation is small (range ≈ 0.159–0.167)
- No clear increasing or decreasing trend
- Filtering proportion remains stable despite changes in cluster granularity

Evaluating cluster coherence with cell type

Method	Concordance with CellTypist	Concordance with singleR
ddqc 0.4	0.8746	0.8764
ddqc 0.8	0.8764	0.8832
ddqc 1.3	0.8791	0.8792
ddqc 1.6	0.8727	0.8728
ddqc 2	0.8715	0.8715
Standard QC	1	1
miQC	0.9754	0.9777

Table 1 Concordance comparison of two reference-based cell type annotation (celltypist and singleR) with multiple resolutions of ddqc, standard QC, and miQC after MAD filtering

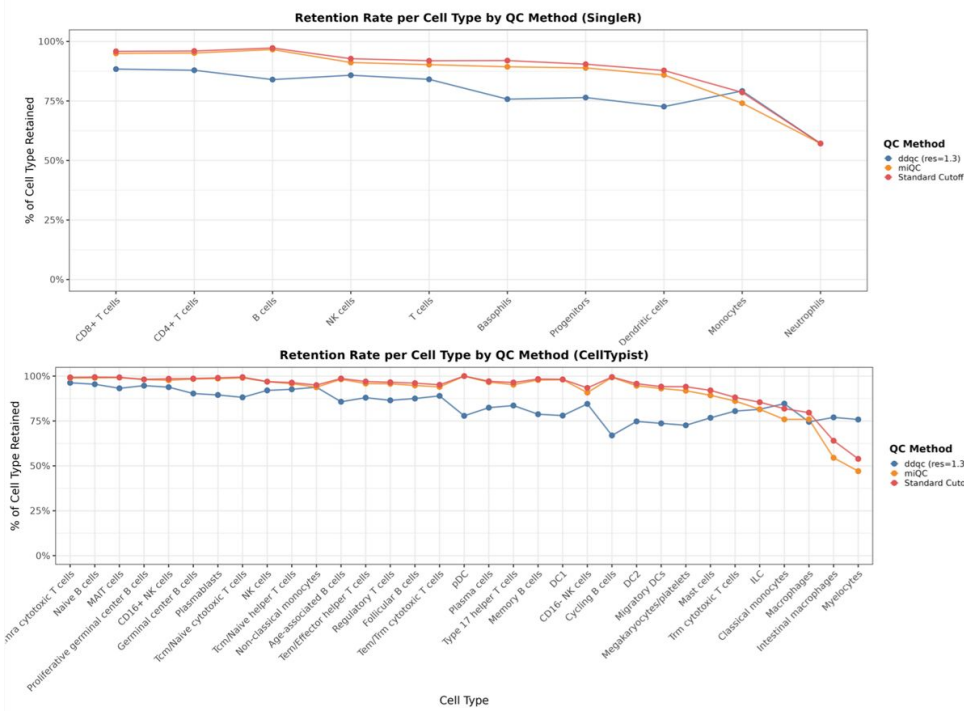
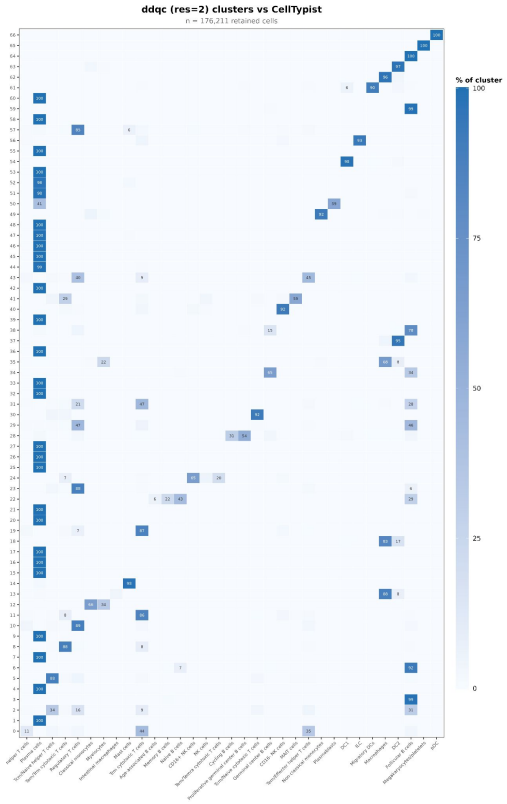
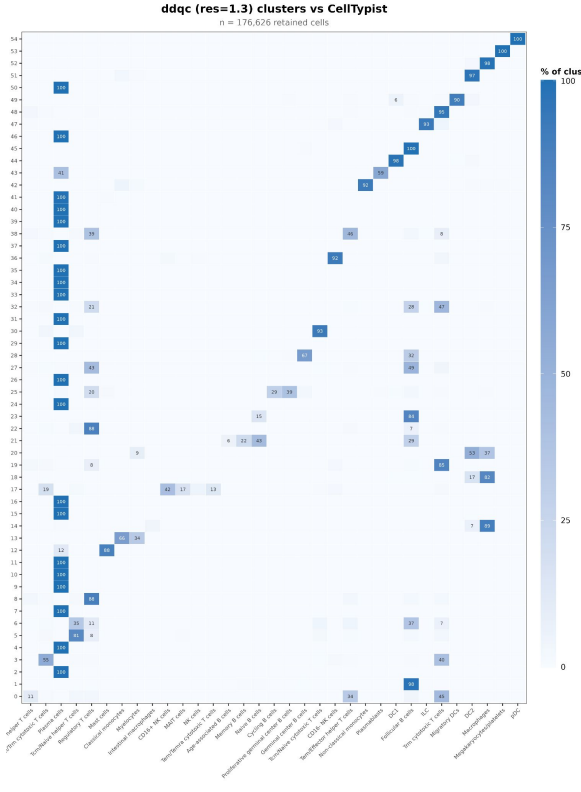


Fig 12 Line graph of retention rate per cell type between ddqc, miQC, and standard cutoff method using singleR (top) and CellTypist (bottom)

Evaluating cluster coherence with cell type





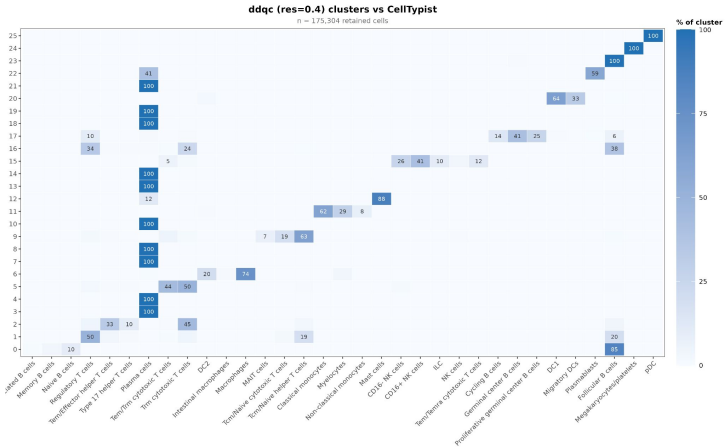


Figure 14. Heatmaps of clustering agreement with reference-based annotation CellTypist